



FIRST SEMESTER, 2026–2027

Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : CS U101
Course Title : Introduction to Python
Instructor-in-Charge : Dr. Prashant Trivedi
Instructors :

1. Course Description:

Introduction to Python, analysis of the problem, design, implementation, and testing, Python programming fundamentals: program structure, data types, variables, expressions, assignment; Program control statements: if, else, logical operators, assignment operators, conditional operators, blocks. Program repetition statements: while, for; Data Structures: Lists, tuples, sets and dictionaries; Using functions: writing functions enhance program modularity; String handling, File handling, Exception handling, Object-oriented fundamentals: classes, objects, methods, abstraction, predefined classes; Writing classes: programmer-defined classes, instance data and methods, parameters, overloading, encapsulation.

2. Scope & Objective of the Course:

Python programming is an essential skill that is increasingly relevant in today's technology- driven world. This course is designed to introduce students to the core concepts of Python programming, guiding them through essential topics such as problem analysis, design, implementation, and testing. Students will gain hands-on experience with Python programming fundamentals, including program structure, data types, variables, and expressions.

The course will cover key programming concepts such as control statements (if, else, logical and conditional operators), repetition statements (while, for loops), and data structures (lists, tuples, sets, and dictionaries). By the end of the course, students will have the knowledge and skills to write Python programs from scratch, design custom classes, and apply core programming techniques to real-world problems.

3. Course learning outcomes

- CLO1.** Explain Python fundamentals including variables, data types, operators, and control structures.
- CLO2.** Develop Python programs using conditional statements, loops, functions, and built-in data structures
- CLO3.** Apply Python to solve basic business problems through calculations, decision logic, and simple automation.

CLO4. Analyze and debug Python programs to identify errors and improve code efficiency and readability.

4. Textbooks:

TB1. Kaefer, F. (2020). Introduction to Python Programming for Business and Science Applications. SAGE Publications, Inc. ISBN 978-1544377445.

References:

Ref1. Matthes, E. (2023). Python Crash Course: A Hands-On, Project-Based Introduction to Programming (3rd ed.). No Starch Press. ISBN 978-171850270

Ref2. Sweigart, A. (2025). Automate the Boring Stuff with Python: Practical Programming for Total Beginners (3rd ed.). No Starch Press. ISBN 978-1718503403.

5. Course Plan:

Table 1: Detailed Weekly Course Plan

Week	Topics to be Covered	Lecture No.	Learning Outcomes	Reference
1	Introduction to Programming & Business Applications of Python; Installing Python, IDEs (Anaconda / VS Code), Running First Program; Python Syntax, Keywords, Comments, Indentation	1–3	CLO1	TB Chapter 1; Lecture Notes/PPTs
2	Variables and Assignment Statements; Data Types: <code>int</code> , <code>float</code> , <code>string</code> , <code>Boolean</code> ; Type Conversion and Input/Output Functions	4–6	CLO1	TB Chapter 2; Lecture Notes/PPTs
3	Operators: Arithmetic, Relational, Logical; Expressions and Operator Precedence; Decision Making: <code>if</code> , <code>if-else</code> statements	7–9	CLO1	TB Chapters 3–4; Lecture Notes/PPTs
4	Nested <code>if</code> and <code>elif</code> ladder; Return Values and Scope of Variables; Looping Concepts: <code>for</code> loop	10–12	CLO2	TB Chapter 4; Lecture Notes/PPTs

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Week	Topics to be Covered	Lecture No.	Learning Outcomes	Reference
5	while loop and loop control statements (break, continue); Introduction to Functions; Introduction to Strings	13–15	CLO2	TB Chapter 5; Lecture Notes/PPTs
6	String Operations and Built-in String Methods; Lists: Creation, Indexing, Slicing; List Methods and Iterations	16–18	CLO2	TB Chapter 6; Lecture Notes/PPTs
7	Tuples and Sets; Dictionaries: Key-Value Pairs and Operations	19–21	CLO2	TB Chapter 7; Lecture Notes/PPTs
8	Data Structures in Business Scenarios; Introduction to NumPy Arrays; Basic Numerical Computations using NumPy	22–24	CLO3	TB Chapter 8; Lecture Notes/PPTs
9	Functions and Modular Programming; Defining and Calling Functions; Parameters and Return Values; Function Decomposition in Solving Business Problems	25–27	CLO3	TB Chapter 9; Lecture Notes/PPTs
10	Introduction to Pandas DataFrames; Data Loading, Selection, and Filtering	28–30	CLO3	TB Chapter 9; Lecture Notes/PPTs
11	Simple Business Data Analysis using Pandas; Iterative Business Calculations using Loops; Business Decision-Making Case Examples	31–33	CLO4	TB Chapter 10; Lecture Notes/PPTs
12	Introduction to Errors and Exceptions; Common Python Errors and Debugging Techniques; Writing Clean and Readable Code (PEP-8 Basics); Code Optimization Basics	34–36	CLO4	TB Chapter 11; Lecture Notes/PPTs

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Week	Topics to be Covered	Lecture No.	Learning Outcomes	Reference
13	Mini Project: Problem Definition & Logic Design; Coding and Implementation; Debugging and Improvement	37–39	CLO4	Mini Project
14	Group Presentation; Final Review, Doubt Clearing & Exam Preparation	40–42	CLO1–CLO4	Course Review

6. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Quiz/Mini project	20%	–	TBA	Open Book
Lab	30%	–	–	Weekly
Mid-Semester	20%	90 mins.	TBA	Closed Book
Comprehensive Examination	30%	180 mins.	TBA	Closed + Open Book

NC Criterion:

- A student should obtain 40% of the median marks of the class to clear the course. If any student gets the marks lower than the prescribed standard mentioned above, he/she will be awarded “NC”.
- A student is likely to get “NC” if he/she doesn’t appear (or appear for the sake of appearing for the evaluation components) or scores zero in the comprehensive exam.
- A student is likely to get “NC” if he/she is scoring zero in the lab component / Abstaining from lab classes throughout.

7. Attendance Policy:

Compulsory in labs (for all weeks) and first week in the classes.

8. Office Consultation Hour:

Will be announced in the class. Also, feel free to email me to seek a prior appointment.

9. Make-up Policy:

Make-up tests will be given only to genuine cases. No make-up will be given for the labs.

Prashant Trivedi
Instructor-in-charge
CS U101